

Mathematical Statistics (I)

Solutions to Midterm Exam.

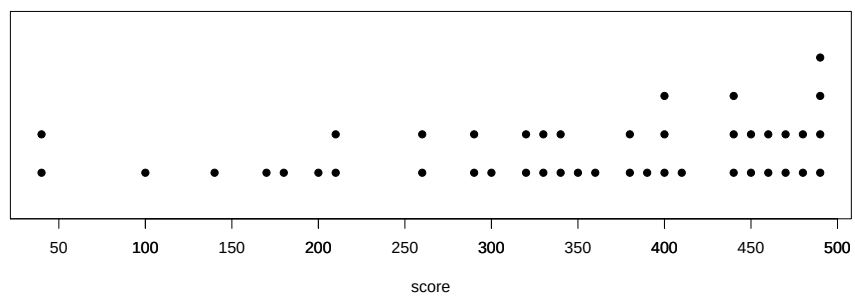
April 18, 2016 (金忠洛 Kim, Choongrak)

- 1.(40) Let X be a r.v. with mgf $M(t) = (1-t)^{-3}, t < 1$. Find $E(X^k)$ for some positive integer k . (Ans.) $M^{(k)}(t) = 3 \cdot 4 \cdots (k+2)(1-t)^{-(k+3)} \Rightarrow E(X^k) = 3 \cdot 4 \cdots (k+2)$.
- 2.(60) Let X be a r.v. with mgf $M(t), t > 0$. Show that $P(X \geq a) \leq e^{-at}M(t)$. (Ans.) By the Markov inequality, we have $P(e^{tX} > e^{ta}) \leq E(e^{tX})/e^{ta}$, i.e., $P(X > a) \leq e^{-at}M(t)$.
- 3.(180) Let (X, Y) have the joint pdf $f(x, y) = 8xyI(0 < x < y < 1)$, and let $Z = X/Y$ and $W = Y$.
- (1)(60) Find the cdf of Z . (Ans.) For $0 < z < 1$, we have $P(Z \leq z) = P(X \leq zY) = \int_0^1 \int_0^{zy} 8xy dx dy = z^2 I(0 < z < 1)$. Also, $F(z) = 1$ if $z > 1$, and $F(z) = 0$ if $z < 0$.
- (2)(40) Using the transformation technique, find the jpdf of (Z, W) . (Ans.) Inverse functions are $x = zw$ and $y = w$, and the Jacobian is $|w|$. Hence, $f(z, w) = 8zw^3 I(0 < w < 1, 0 < z < 1)$.
- (3)(40) Are X and Y independent? Why or why not? (Ans.) No, because we cannot express $f(x, y)$ as a product of two functions; $g(x)$ and $h(y)$, say.
- (4)(40) Are Z and W independent? Why or why not? (Ans.) Yes, because $f(z, w) = 8z I(0 < z < 1) \cdot w^3 I(0 < w < 1)$.
- 4.(60) Let $f(x, y) = (x + y)I(0 < x < 1)I(0 < y < 1)$ be the jpdf of (X, Y) . Compute the correlation coefficient between X and Y . (Ans.) $E(X) = E(Y) = 7/12, Var(X) = Var(Y) = 11/144, Cov(X, Y) = -1/144 \Rightarrow Corr(X, Y) = -1/11$.
- 5.(160) Let (X, Y) have the joint pdf $f(x, y) = 2I(0 < x < y < 1)$.
- (1)(40) Compute $E(X|Y)$. (Ans.) $f(y) = \int_0^y 2dx = 2yI(0 < y < 1) \Rightarrow f(x|y) = f(x, y)/f(y) = y^{-1}I(0 < y < 1) \Rightarrow E(X|Y) = \int_0^y xy^{-1}dx = \frac{y}{2}I(0 < y < 1)$.
- (2)(40) Compute $Var(E(X|Y))$. (Ans.) $Var[E(X|Y)] = E(Y^4/4) - E^2(Y/2) = 1/8 - 1/9 = 1/72$.
- (3)(40) Compute $Var(X|Y)$. (Ans.) $Var(X|Y) = E(X^2|Y) - E^2(X|Y) = \int_0^y x^2 y^{-1}dx - y^2/4 = \frac{1}{12}y^2 \cdot I(0 < y < 1)$.
- (4)(40) Compute $E(Var(X|Y))$. (Ans.) $E(Var(X|Y)) = E(Y^2/12) = \int_0^1 y^2 \cdot 2y/12 dy = 1/24$.

Summary Statistics

$n = 44$, average = 344

Dot Diagram of Scores



Top Seat : 김용갑, 박수진, 안민지, 정이선